

● HARD

● GEOMETRY AND TRIGONOMETRY

Circles

5 questions · ~8 min

04

Q1

GEOMETRY AND TRIGONOMETRY

$$x + 4^2 + y - 19^2 = 121$$

The graph of the given equation is a circle in the xy -plane. The point a, b lies on the circle. Which of the following is a possible value for a ?

- A. -16
- B. -14
- C. 11
- D. 19

The equation $x^2 + y - 2^2 = 36$ represents circle A. Circle B is obtained by shifting circle A down 4 units in the xy -plane. Which of the following equations represents circle B?

A. $x^2 + y + 2^2 = 36$

B. $x^2 + y - 6^2 = 36$

C. $x - 4^2 + y - 2^2 = 36$

D. $x + 4^2 + y - 2^2 = 36$

Q3

GRID-IN

GEOMETRY AND TRIGONOMETRY

A circle in the xy -plane has a diameter with endpoints $2, 4$ and $2, 14$. An equation of this circle is $x^2 + y^2 - 9^2 = r^2$, where r is a positive constant. What is the value of r ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Circle A has equation $x - 7^2 + y + 3^2 = 1$. In the xy -plane, circle B is obtained by translating circle A to the right 4 units. Which equation represents circle B?

A. $x - 7^2 + y + 7^2 = 1$

B. $x - 3^2 + y + 3^2 = 1$

C. $x - 11^2 + y + 3^2 = 1$

D. $x - 7^2 + y - 1^2 = 1$

Q5

GRID-IN

GEOMETRY AND TRIGONOMETRY

Points A and B lie on a circle with radius 1, and arc $\overline{AB}$ has length $\frac{\pi}{3}$. What fraction of the circumference of the circle is the length of arc $\overline{AB}$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.